Dialogue as a Coupled Neural System

Interpersonal Synchronization, Shared Reference Frames, and Informative Divergence

Micah Blumberg
Self Aware Networks Research Institute
selfawarenetworks.com

Preprint — August 13, 2026

Abstract

Two people solving a problem together must become similar enough to know that they are discussing the same object, event, question, or goal, yet remain different enough to contribute independent evidence, identify errors, and explore alternatives. Research on speaker-listener coupling, cooperation, face-to-face dialogue, turn-taking, classroom interaction, symbolic communication, and natural conversation has shown reliable relations among participants' neural and behavioral signals. Those relations are scientifically important, but they do not all mean the same thing. Shared stimulation, matched speech envelopes, coordinated movement, physiological covariation, neural synchrony, representational alignment, reciprocal information transfer, comprehension, and task success require separate measurements and controls.

This paper develops a source-faithful Self Aware Networks (SAN) framework for dialogue as an environment-mediated coupled neural system. Its genealogy begins with Micah Blumberg's 2012 descriptions of contingent EEG feedback as a two-way conversation, sensory-to-electrical communication across people, synchronous and asynchronous linkage, and multiscale collective intelligence. A 2021 source family sharpened the problem through the broccoli example: different brains can recognize and discuss the same object without carrying identical microscopic encodings, so successful translation must preserve task-relevant relations. June 2022 SAN notes then stated that coherence can support coupling while remaining insufficient for distinction. The controlled mathematical and experimental program presented here is a 2026 synthesis and is not backdated.

The framework models each participant's private evidence, learned receiver state, messages, shared public history, reference state, neural measurements, and task outcome. It separates raw synchrony from representational alignment and lagged conditional influence, then decomposes task-relevant information into redundant, unique, and synergistic parts. Within each participant, a receiver-relative departure from expected tonic activity may be represented as a Phase Wave Differential (PWD). Across people, the PWD does not cross directly between brains: speech, gaze, gesture, action, and shared artifacts transduce one participant's state into sensory consequences for another. The paper specifies a Dialogue Coupling Benchmark, a prospective dual-EEG and dual-fNIRS protocol, a machine-checked finite protocol kernel, and a completed controlled pilot with two fixed Qwen2.5 instruction-tuned models. In that pilot, the two preregistered superiority endpoints failed: current dialogue did not beat isolation on complementary-object decisions, and its favorable current-over-stale correction difference missed the frozen effect and multiplicity criteria. Exact replay preserved every live prediction. The result is retained as a bounded negative finding; it is not human or neural evidence.

1. Start with an ordinary problem

Imagine that two people are trying to navigate a damaged building. One person has a map showing which hallways remain open. The other has a sensor report showing where smoke is spreading. Neither person can choose a safe route alone. They have to establish that "the east stairwell," "the red room," and "the second turn" refer to the same places. They have to time questions and responses so that a correction arrives before a decision is made. They also have to preserve what only one of them knows. If the map holder simply copies the sensor holder's opinion, the route information is lost. If the sensor holder refuses to update, the two private models never become useful together.

Successful dialogue therefore requires more than agreement. It requires a changing balance among shared reference, prediction, independent evidence, correction, confidence, and action. A useful scientific account should be able to answer at least six questions:

1. What information did each participant possess before speaking?
2. Which public signal carried that information into the shared environment?
3. What did the receiver actually detect and represent?
4. Did the receiver's later state or behavior depend on the sender after common inputs were controlled?
5. Which similarities reflected useful common ground, and which merely reflected shared timing or movement?
6. Did convergence preserve or erase the differences needed to solve the task?

Only after these distinctions are in place is it useful to name neural synchrony, interbrain coupling, representational alignment, common ground, informative divergence, or a coupled neural system.

2. The physical loop between separate nervous systems

A conversation is not a direct wire between brains. Participant A's neural and bodily state contributes to speech, gaze, expression, gesture, writing, or manipulation of a shared object. Those physical events change participant B's sensory input. B's nervous system transforms that input according to B's anatomy, learned history, current attention, affect, expectations, and goals. B then produces a response that returns through the environment and changes A.

Let participant i have internal state $X_i(t)$, private evidence E_i, observation $O_i(t)$, emitted message or action $M_i(t)$, and receiver parameters $\Theta_i(t)$. Let $W(t)$ be the shared-world state and $H(t)$ the public interaction history. A minimal update is

$$X_i(t+1) = F_i(X_i(t), E_i, O_i(t), H(t); \Theta_i(t)). \tag{1}$$

The participant emits a public event through an encoding-and-action operation,

$$M_i(t) = \mathcal{E}_i(X_i(t), E_i, H(t)), \tag{2}$$

and the environment transforms that event before it reaches the partner,

$$O_j(t+\tau) = \mathcal{T}_W(M_i(t), W(t), \eta(t)), \tag{3}$$

where τ includes production, propagation, sensing, and processing delays and η includes noise and unmeasured influences. Equations 1-3 make the physical claim modest and testable: interpersonal coupling is mediated by a channel through the shared world. Nothing in this model requires a literal field, spike train, or phase wave to pass directly from one skull to another.

The loop becomes reciprocal when B's changed state affects a later message that changes A:

$$X_i(t) \to M_i(t) \to O_j(t+\tau_{ij}) \to X_j(t+\tau_{ij}+1) \to M_j(t') \to X_i(t'+\tau_{ji}+1). \quad (4)$$

This returned consequence distinguishes a dialogue from a one-way broadcast. It also explains why playback, scripted speech, delayed feedback, and shuffled partners are necessary controls rather than optional embellishments.

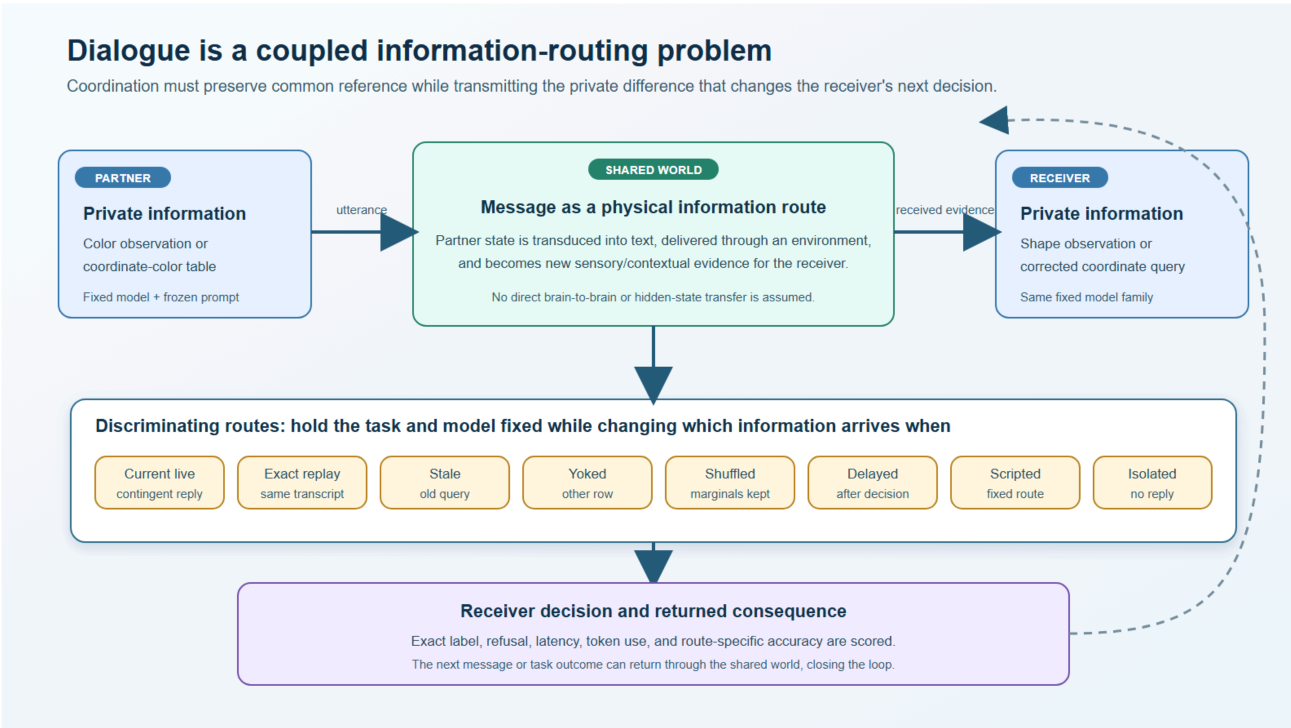

Figure 1. Environment-mediated dialogue and its discriminating routes. The diagram states the physical and computational claim without direct hidden-state transfer between participants.

3. What established research has measured

Speaker-listener fMRI work showed that a speaker's spatiotemporal activity could predict listener activity, with both delayed mirroring and anticipatory responses, and that coupling related to comprehension [2]. Cooperation studies using near-infrared spectroscopy reported increased interpersonal coherence in superior frontal cortex during cooperation and an association with performance [3]. Face-to-face dialogue produced left inferior frontal synchronization relative to several monologue and orientation controls, with multimodal interaction and turn-taking contributing to the effect [4]. EEG and MEG studies have measured interbrain relations during coordinated speech rhythms and verbal turn-taking [5,9]. Classroom and naturalistic-interaction studies have connected synchrony with engagement, affiliation, gaze, affect, or behavioral coordination [7,8].

These findings establish a serious interpersonal neuroscience program. They also establish why one score is insufficient. Synchrony can arise because both people hear the same sound, look at the same event, move at the same tempo, breathe similarly, or occupy corresponding task phases. Burgess distinguished reciprocal, common-driver, one-way-driven, and coincidental synchronization and showed that real condition structure can generate misleading hyperconnections [6]. More recent simulations and EEG analyses demonstrate sensitivity to estimator choice, epoch length, signal-to-noise ratio, and temporal decisions [13].

Content alignment is another variable. Varlet and Grootswagers showed in simulation that conventional interbrain synchrony can remain relatively insensitive to whether two people represent the same or different objects, whereas representational analysis can recover information alignment [12]. Representational similarity methods compare the geometry of response patterns rather than only their simultaneous amplitude or phase [14,15]. A dyad can therefore be synchronized in time while representing different content, or represent corresponding content without maintaining one fixed phase relation.

Causal evidence is beginning to separate these layers further. In a novel symbolic communication task, Liu and colleagues combined behavior, fNIRS hyperscanning, and in-phase, anti-phase, and sham hyper-tACS. Shared intentionality, communicative accuracy, and right superior temporal synchronization were related during system formation, and the stimulation manipulation supported a causal role in that preparation [10]. The effect was stage specific: once the communication system was established, the same variables no longer played the same role. That result is especially important for the present framework because it rejects the assumption that a coupling variable has one fixed function throughout learning.

Natural dialogue also shows the value of divergence. Speer and colleagues found that friends began more aligned and then explored more divergent neural, linguistic, and topic trajectories, whereas strangers tended to converge while building common ground. Greater exploration could relate to a better conversation [11]. Collective-decision studies likewise show that communication can improve calibration and performance under some conditions [16,17], while social influence can reduce diversity and crowd accuracy [18]. In hidden-profile tasks, dissent can improve decision quality by increasing discussion of otherwise unshared evidence [19], and formal work on group integration shows why the combination rule matters [20]. The question is not whether synchrony or divergence is good. The question is which relation preserves and transmits the information needed for the current task.

Two 2025 studies sharpen the bridge from timing to shared models and content without collapsing those variables. Koike and colleagues used a reciprocal maze task in which each partner possessed only part of the solution; partner- and task-specific default-mode-network synchronization was reported during real-time conversation [21]. Carollo and colleagues related separately coded emotional and syntactic/semantic dialogue features to prefrontal fNIRS synchrony in 42 dyads, with different explanatory patterns at global and regional levels [22]. These are preparation-bound results. They motivate explicit content, partner, task, and model controls; they do not turn one synchrony score into a universal measure of shared meaning.

4. The SAN genealogy of the problem

The historical SAN line reached this question through several routes that should remain separate from the 2026 formalism.

In a public 2012 Neo Mind Cycle dialogue, contingent EEG-driven light and sound were contrasted with one-way entrainment and described as a “two-way conversation.” The same source linked simultaneous visual, auditory, and felt activity with learned coordination and treated spoken interaction as communication among larger populations of neurons. In *Chicken Cloud*, the source distinguished neurons that fire together from neurons that fire out of sync, described observed movement and speech as sensory patterns converted into electrical signals in the receiver, and used inhibition to explain why internally modeled action need not become self-action. The *Inner Peace Dialog* added a multiscale analogy: different people can contribute to collective intelligence because sensory communication is transformed back into neural electrical patterns, even though the participants remain distinct.

These sources do not establish modern hyperscanning results in advance. Their value is the problem architecture: reciprocal feedback, sensory transduction, multiscale communication, and preserved distinction.

The 2021 source family sharpened the representational problem. The broccoli example asks how two people can both know and discuss broccoli even if the temporal-spatial patterns inside their brains are not microscopically identical. A direct brain-computer interface cannot assume that copying a raw activity pattern from one person into another will preserve meaning. It may require a third, task-grounded representation that preserves relations among object properties, context, report, action, and each person's learned mapping. A separate 2021 dialogue source describes conversation as participant modeling, inferred goals, listening, affective appraisal, turn-taking, and a decision about whether collaboration continues.

Two public Git notes from 2022 add the central distinction. `a0142z.md` states that synchronization or coherence can support binding, convergence, and coupling while remaining insufficient for distinction, deconvergence, and decoupling. `b0007y.md` proposes that communication infrastructure can magnify collective intelligence through the number and rate of connections, while the later source review correctly preserves the need for routing and transformation rather than connectivity alone. These are direct ancestors of the paper's thesis: coordination supplies a common context, but useful cognition also requires structured difference.

SAOv9 makes “brainwave synchronization through dialog,” group communication, collective intelligence, artificial-agent dialogue, ethics, and groupthink an explicit research agenda. SAOv10 and its governed atom adjudication add the exact control structure used here: lag, direction, common input, speech, motion, familiarity, task success, shared reference, informative divergence, artificial comparators, privacy, manipulation, and the right to retain cognitive difference. Those detailed controlled formulations are 2026 synthesis.

4.1 Exact provenance boundary

The public 2012 sources are bound to three separate records rather than summarized as one retrospective memory. The Neo Mind Cycle/Happiness Makeover transcript is tied to public video `jtj_J6hiL9o`, platform time 2012-08-01T20:05:48-07:00, source lines 81-103 for the contingent two-way feedback argument, and lines 273-291 for sensory transduction across interacting people. *Chicken Cloud* is tied to `uA-TDPycm0g`, source lines 69-103 for firing-together/firing-apart, inhibition, movement, speech, and the interbrain functional analogy. *Inner Peace Dialog* is tied to `QX-0ef31tAU`, platform time 2012-08-23T21:15:59-07:00, and lines 93-97 for the collective-intelligence and sensory-to-electrical conversion argument. File hashes, speaker boundaries, date classes, and custody limits are frozen in the companion source-locator audit.

The preserved text exports do not provide sentence-level audio timecodes for those three passages. This paper therefore cites exact transcript lines plus the public recording identity and platform time; it does not invent audio timestamps. The 2021 recordings retain event dates and later Git custody as separate facts, remain paraphrase-only where speaker attribution is not yet diarized, and are not presented as public 2021 proof. The 2022 and 2024 claims are tied to immutable Git commits and blob identifiers. The 2026 equations, benchmark, estimator choices, and controlled interpretation remain current synthesis.

5. A typed measurement taxonomy

Let $C(t)$ collect common causes visible to both participants, including task events and shared stimuli. Let $A_i(t)$ denote speech acoustics, $B_i(t)$ visible behavior and movement, $P_i(t)$ physiology, and $N_i(t)$ measured neural activity. A raw interbrain relation might be written

$$S_{ij}(f,t) = \left|\mathbb{E}\left[e^{\mathrm{i}(\phi_i(f,t)-\phi_j(f,t))}\right]\right|, \tag{5}$$

but Equation 5 is only a phase-relation statistic. It does not say why the relation exists or what content it carries, and any electrophysiological phase estimator must retain volume-conduction, noise, and sample-size controls [29].

The first repair is nuisance-conditioned neural residualization,

$$\widetilde{N}_i(t) = N_i(t) - \widehat{\mathbb{E}}[N_i(t) \mid C(t), A_i(t), B_i(t), P_i(t), Q(t)], \tag{6}$$

where $Q(t)$ contains device, quality, and task-phase covariates. Residualization does not magically recover causation; it makes the conditioning assumptions explicit. Equation 6 is a cross-fitted diagnostic baseline, not the confirmatory causal estimator. Speech, gaze, and movement may be mediators of dialogue rather than pretreatment nuisances. Conditioning on them estimates a residual or direct path and cannot be described as the total effect of the public communication channel.

A lagged observational directional relation can then be defined as conditional information. Let $Z_{ij}^{\mathrm{pre}}(t,\tau)$ contain only the frozen common causes and quality variables measured before the candidate sender event, including shared stimulus, task phase, turn assignment, and target self-history:

$$T_{i\to j}^{\mathrm{obs}}(\tau) = I\Big(N_i^{<t}; N_j(t+\tau) \mid N_j^{<t}, Z_{ij}^{\mathrm{pre}}(t,\tau)\Big). \tag{7}$$

Equation 7 is a predictive-dependence model family, not an intervention effect. Transfer entropy, conditional mutual information, state-space prediction, and causal modeling make different assumptions [23,24,32-34]. Unmeasured common causes can keep $T_{i\to j}^{\mathrm{obs}}$ positive even after extensive conditioning. The selected estimator must be validated on synthetic fixtures and compared with time-shifted, shared-phase, and shuffled-pair nulls. A causal statement additionally requires a declared structural model or randomized channel intervention.

Representational alignment requires a declared set of objects, meanings, states, or task conditions. Let $D_i(t)$ be participant i's cross-validated representational dissimilarity matrix. Then

$$R_{ij}(t) = \mathrm{corr}_{\mathrm{cv}}(\mathrm{vec}\, D_i(t), \mathrm{vec}\, D_j(t)) \tag{8}$$

measures alignment of representational geometry. It must remain separate from S_{ij} and $T_{i\to j}$.

The public reference state $G(t)$ records which entities, properties, and relations have been mutually established:

$$G(t+1) = \mathcal{U}(G(t), M_i(t), M_j(t), W(t), \epsilon_G(t)). \tag{9}$$

Reference accuracy is evaluated against the task's immutable world model,

$$A_G(t) = 1 - d(G(t), G^\star(t)), \tag{10}$$

with d defined for the task rather than chosen after seeing the result.

6. Common ground is not enough

Let Y be the correct decision or task-relevant latent state. Let Z_i and Z_j be the information available from each participant after measurement and encoding. A partial-information decomposition represents

$$I(Y; Z_i, Z_j) = I_{\text{red}} + I_{\text{uniq},i} + I_{\text{uniq},j} + I_{\text{syn}}, \tag{11}$$

where the terms denote redundant information, unique information from each partner, and information available only from their joint configuration. Shannon's information measure supplies the common unit, bits when logarithms use base 2 [1]. The bivariate BROJA construction is the confirmatory discrete-task decomposition, while Williams-Beer $I_{\min}$ and pointwise common-surprisal decomposition remain sensitivity analyses [25-27]. Equation 11 is the conceptual contract; Equations 27-30 below freeze the operational definition.

A dialogue that maximizes shared agreement while eliminating I_{uniq} can become confidently wrong. A dialogue that preserves every difference without establishing $G(t)$ can fail to coordinate. The proposed objective therefore contains both alignment and retention:

$$J = \alpha A_Y + \beta A_G + \gamma C_{\text{err}} + \delta\, \text{Cal} + \mu I_{\text{syn}} + \nu(I_{\text{uniq},i} + I_{\text{uniq},j}) - \rho L_{\text{uniq}} - \kappa K, \tag{12}$$

where A_Y is task accuracy, C_{err} correction performance, Cal confidence calibration, L_{uniq} loss of valid unique evidence, and K time or resource cost. The coefficients are not universal constants. They express a task-specific multi-objective problem.

An informative-divergence score can be written

$$D_{ij}^{+}(t) = D_{ij}(t)\left[U_i^{(G)} + U_j^{(G)}\right], \tag{13}$$

where raw difference D_{ij} receives value only when it preserves task-relevant unique information. The terms $U_i^{(G)}$ and $U_j^{(G)}$ are the task-state-conditioned BROJA unique-information quantities defined below. They are not literal subtraction of random variables. Irrelevant noise can increase difference without increasing D^{+}. This prevents “divergence is good” from replacing “synchrony is good” as a new simplification.

7. From local neural difference to interpersonal consequence

SAN proposes that a receiving neural population is already in a learned and ongoing state before a message arrives. Let $\Theta_{i,r}(t)$ denote the slower learned parameters of receiver r in participant i, and let $\widehat{u}_{i,r}(t)$ be its expected tonic relation under current context. For measured event vector $u_{i,r}(t)$, define a receiver-relative departure

$$\delta_{i,r}(t) = u_{i,r}(t) - \widehat{u}_{i,r}(t \mid X_i^{<t}, \Theta_{i,r}, C_i). \tag{14}$$

For a discrete event, conditional surprisal uses a probability mass. For a continuous departure, a density alone is dimensional and coordinate dependent. The protocol therefore freezes, using training data only, a finite measurable partition $\mathcal{B} = \{B_1, \ldots, B_K\}$ of the declared feature space. If $\delta_{i,r}(t) \in B_k$, the event's conditional surprisal is

$$\mathcal{I}_{i,r}(t) = -\log_2 P\big(\delta_{i,r}(t) \in B_k \mid X_i^{<t}, \Theta_{i,r}, C_i\big). \tag{15}$$

The feature transformation, partition boundaries, support, and smoothing rule are frozen from training data. Values are interpreted only within that declared representation; changing the partition or coordinates defines a different estimand. A density-based differential-surprisal analysis may be reported separately, but it cannot inherit Equation 15's coordinate-invariant probability-mass interpretation.

A departure matters biologically only if the tuned receiver transforms it into a consequence:

$$q_{i,r}(t) = \Psi_{i,r}(\delta_{i,r}(t), \mathcal{I}_{i,r}(t), \Theta_{i,r}(t), \text{inhibition}, \text{neuromodulation}). \tag{16}$$

This candidate receiver-relative operation is called a Phase Wave Differential when phase and waveform relations are among the defined components. It is not a synonym for phase coding, traveling waves, coherence, or prediction error. It names the typed difference among an arriving event, the receiver's expected relation, the receiver's learned state, and the downstream consequence.

Dialogue links two local operations through physical transduction:

$$q_{i,r}(t) \to M_i(t) \to O_j(t+\tau) \to \delta_{j,s}(t+\tau) \to q_{j,s}(t+\tau) \to M_j(t'). \tag{17}$$

Equation 17 is the paper's central SAN chain. It does not say that participant A's PWD becomes participant B's identical PWD. It predicts that a consequential local difference can alter a public event, which a differently tuned receiver transforms into another local difference and a returned action.

The incremental test is whether PWD features improve held-out prediction beyond ordinary alternatives:

$$\Delta_{\text{PWD}} = \text{Perf}(\mathcal{M}_{\text{base+PWD}}) - \text{Perf}(\mathcal{M}_{\text{base}}), \tag{18}$$

where performance is converted in advance to an oriented scale on which larger is better, and the baseline includes rate, power, raw phase, speech envelope, movement, physiology, self-history, partner history, and representational state. A useful result should also survive a selective perturbation and matched rescue.

8. The Dialogue Coupling Benchmark

The proposed benchmark uses tasks with known evidence allocation. In complementary trials, each partner holds information the other lacks. In redundant trials, both receive the same decisive clue. In conflicting trials, cues differ in reliability. In reference-learning trials, partners must establish names for unfamiliar objects or spatial relations. In correction trials, one participant receives a planted false premise and the other receives disconfirming evidence.

Each task is crossed with solo, silent shared-stimulus, scripted exchange, yoked playback, delayed dialogue, live reciprocal dialogue, shuffled-pair analysis, reference remapping, and exact rescue conditions. Audio, gaze, movement, confidence, task events, and outcomes are synchronized with dual EEG or, in a separately justified slower design, fNIRS. The modalities are not treated as interchangeable. EEG records electrical consequences of population currents with high temporal but limited source precision; fNIRS measures slower hemodynamic changes and requires explicit systemic-physiology controls [30]. Each requires its own forward model and confounds.

The model ladder is

$$\mathcal{M}_0 \subset \mathcal{M}_1 \subset \mathcal{M}_2 \subset \mathcal{M}_3 \subset \mathcal{M}_4 \subset \mathcal{M}_5 \subset \mathcal{M}_6, \tag{19}$$

from individual ability; to shared input and behavior; to raw interbrain measures; to representational alignment; to directional conditional influence; to common-ground/PID structure; to SAN receiver-relative features. Model complexity and information access are reported, and evaluation occurs on held-out dyads and task instances.

Pair specificity is measured by comparison with shuffled partners,

$$\Delta_{\text{pair}} = T^{\text{obs,true}}_{i \to j} - \mathbb{E}_{j' \neq j} \left[T^{\text{obs,shuffled}}_{i \to j'} \right], \tag{20}$$

and reciprocal benefit by comparison with yoked input,

$$\Delta_{\text{recip}} = \text{Perf}(\text{live reciprocal}) - \text{Perf}(\text{matched yoked playback}). \tag{21}$$

The first causal perturbation changes the communication channel rather than the nervous system. A calibrated delay or reference-token remapping should selectively impair partner prediction, reference accuracy, or correction if the loop depends on that relation. Restoring the exact latency or mapping supplies a matched rescue:

$$\mathcal{R} = \frac{Y_{\text{rescue}} - Y_{\text{perturb}}}{Y_{\text{baseline}} - Y_{\text{perturb}}}, \qquad Y_{\text{baseline}} - Y_{\text{perturb}} > \delta_{\text{min}} > 0. \tag{22}$$

9. Hypotheses

H1 — Reciprocal direction. Live dialogue will produce pair-specific lagged conditional information beyond shared stimulus, speech, movement, physiology, self-history, yoked playback, and shuffled-pair nulls.

H2 — Content over timing. Cross-validated representational alignment will predict reference accuracy and comprehension beyond raw amplitude or phase synchrony.

H3 — Complementary evidence. When partners possess unique valid evidence, the best-performing dyads will preserve more task-relevant unique information than matched high-convergence dyads that suppress one partner's clue.

H4 — Task-dependent balance. The relation between convergence and performance will be nonmonotonic and will shift with evidence redundancy, conflict, relationship, and task phase.

H5 — Formation versus maintenance. Coupling measures that predict formation of a new reference system will not necessarily remain predictive after that system becomes familiar.

H6 — Receiver-relative consequence. Local tonic-relative, receiver-conditioned features will add held-out prediction of correction, turn transition, or action beyond rate, power, raw phase, waveform, and history baselines.

H7 — Perturbation and rescue. Calibrated delay or reference remapping will selectively degrade directional update and task performance, and exact restoration will selectively recover them.

H8 — Typed human/artificial correspondence. Artificial agents with persistent shared context and preserved private evidence will outperform isolated or summary-only agents on reference tracking and correction, but software-state similarity will not be equated with electrophysiological synchrony.

For a preregistered endpoint (Y_k), a bounded hypothesis test is

$$H_{0,k} : \Delta_k \leq \epsilon_k, \qquad H_{1,k} : \Delta_k > \epsilon_k, \tag{23}$$

where ϵ_k is a declared smallest effect of interest. Failure to exceed ϵ_k refines the named operational hypothesis in that preparation. It is not converted into a global classification about SAN and science.

10. Artificial-agent application

The same evidence-allocation tasks can be run with artificial agents under four conditions: isolated state, one-shot summary exchange, full message exchange with separate state, and versioned persistent shared context with preserved private evidence. The comparison measures object identity, reference errors, provenance, correction latency, confidence calibration, unique-evidence retention, and final accuracy.

Let $K_i(t)$ be private agent state and $G_A(t)$ versioned shared state. A persistent-context update is

$$(G_A(t+1), v_{t+1}) = \mathrm{Commit}(G_A(t), v_t, \Delta_i(t), \pi_i), \tag{24}$$

where v_t is the expected version, Δ_i a proposed change, and π_i the actor and provenance record. Unique private evidence remains separately addressable rather than being overwritten by consensus.

The human/artificial comparison is deliberately typed. Both systems can share object identity, turn state, commitments, uncertainty, and correction history. Only biological participants supply electrophysiological measurements, embodied interoception, and neural receiver dynamics. Functional correspondence does not erase substrate or mechanism differences.

10.1 Mathematical and estimator contract

The formal objects now have declared domains. Time is represented on a synchronized grid $t \in \{0, \Delta t, 2\Delta t, \ldots\}$ in seconds; modality-specific feature windows retain their start, stop, center, and sampling rate. Neural feature vectors satisfy $N_i(t) \in \mathbb{R}^{p_N}$ after a frozen preprocessing pipeline. Acoustic, behavioral, physiological, quality, and common-driver covariates occupy separately versioned real or categorical feature spaces. Messages $M_i(t)$ are timestamped events with text, acoustic, referential, and action fields; they are not assumed to share units with neural data. $G(t)$ is a typed graph over immutable task objects and relations. Outcome Y is declared per task as categorical, ordinal, count, duration, or bounded continuous before analysis.

Units and ranges are not left implicit. S_{ij} is dimensionless in $[0, 1]$; R_{ij} is a dimensionless cross-validated geometry correlation in $[-1, 1]$ when correlation is the frozen comparator; and A_G is dimensionless in $[0, 1]$ only when d is preregistered and normalized to $[0, 1]$. Information, surprisal, transfer, redundancy, unique information, and synergy use base-2 logarithms and are reported in bits per declared event or window; a bits-per-second rate is a separate estimand. Continuous departures use Equation 15's training-frozen event partition; values cannot be compared after changing feature coordinates, partition boundaries, support, or smoothing. The receiver departure $\delta_{i,r}$ retains the units of its frozen feature vector, while $q_{i,r}$ and performance inherit the declared endpoint units. D^+ is in bits only when D_{ij} is dimensionless. Before Equation 12 is computed, every endpoint is transformed to a declared dimensionless utility scale; otherwise the coefficients must carry compensating units and J cannot be compared across tasks. The rescue ratio is dimensionless and is defined only when the oriented baseline-minus-perturbation contrast exceeds $\delta_{\min}$, which has the outcome's units. No additive stabilizer is used to manufacture a defined ratio.

For a preregistered lag set $\mathcal{L}$ fixed from the task and modality, directional conditional information is estimated only at lags in that set:

$$\widehat{T}^{\mathrm{obs}}_{i \to j} = \max_{\tau \in \mathcal{L}_{\mathrm{train}}} \widehat{I}_{\mathrm{KSG}}\Big(N_i^{<t}; N_j(t+\tau) \mid N_j^{<t}, Z_{ij}^{\mathrm{pre}}(t, \tau)\Big), \tag{25}$$

with lag selection confined to training folds and the selected lag evaluated unchanged on held-out dyads. The estimator is a conditional k-nearest-neighbor method grounded in transfer-entropy, mutual-information, and conditional-mutual-information work [23,24,35]. Natural-log estimator outputs are divided by $\ln 2$ before reporting bits. Discrete low-count tasks use a bias-corrected discrete estimator. No directional estimate is interpreted when effective sample size, neighbor occupancy, stationarity, or surrogate calibration fails its preregistered threshold.

Representational alignment uses a cross-validated distance estimator rather than an in-sample correlation of noisy means. For task conditions a and b,

$$d_i(a,b) = \left(\widehat{\mu}_{i,a}^{(A)} - \widehat{\mu}_{i,b}^{(A)}\right)^\top \widehat{\Sigma}_{i,\text{train}}^{-1} \left(\widehat{\mu}_{i,a}^{(B)} - \widehat{\mu}_{i,b}^{(B)}\right), \tag{26}$$

where A and B are independent folds and $\widehat{\Sigma}^{-1}$ is estimated from training data [28]. Negative cross-validated distances remain possible and are not clipped. Cross-participant geometry is compared only after object labels, folds, feature dimension, and distance normalization are frozen.

For finite discrete target Y, sources Z_i, Z_j, and a preregistered finite public-reference stratum $G = g$, let $P_g = P(Y, Z_i, Z_j \mid G = g)$ and define the BROJA constraint set

$$\Delta_{P_g} = \{Q_g : Q_g(Y, Z_i) = P_g(Y, Z_i),\ Q_g(Y, Z_j) = P_g(Y, Z_j)\}, \tag{27}$$

and the two public-state-conditioned unique-information terms

$$U_i^{(G)} = \sum_g P(g) \min_{Q_g \in \Delta_{P_g}} I_{Q_g}(Y; Z_i \mid Z_j), \qquad U_j^{(G)} = \sum_g P(g) \min_{Q_g \in \Delta_{P_g}} I_{Q_g}(Y; Z_j \mid Z_i). \tag{28}$$

Conditional redundancy and synergy are then

$$R^{(G)} = I_P(Y; Z_i \mid G) - U_i^{(G)} = I_P(Y; Z_j \mid G) - U_j^{(G)}, \tag{29}$$

$$S^{(G)} = I_P(Y; Z_i, Z_j \mid G) - R^{(G)} - U_i^{(G)} - U_j^{(G)}. \tag{30}$$

These definitions close the earlier ambiguity for the confirmatory benchmark [26]. They are computed only when each declared $G = g$ stratum passes a preregistered sample-size and support threshold; otherwise the conditional PID endpoint is not reported. The analysis must still report sensitivity to at least one justified alternative decomposition [25,27], optimization tolerance, bootstrap uncertainty, and finite-sample bias. If the decompositions disagree in sign or ordering for a conclusion, that conclusion remains decomposition-dependent.

The rescue ratio in Equation 22 is reported only when the baseline-minus-perturbation contrast exceeds a preregistered minimum magnitude $\delta_{\text{min}} > 0$. Otherwise the ratio is undefined and the raw paired contrasts are reported. Bounded outcomes use an identity, logit, ordinal, count, duration, or survival model appropriate to their measurement scale; they are not silently combined. Dyad and participant effects are modeled separately, with crossed or nested random effects declared from the design [31]. Missing samples carry reason codes and are never interpolated across turn boundaries, perturbations, or condition changes. Multiple imputation, if used for covariates, is fitted within training data and never applied to the primary neural time series.

The identifiability contract includes six required counterexamples: a common driver producing apparent synchrony without reciprocity; acoustically matched speech with different represented content; aligned content at asynchronous times; synchronized activity carrying irrelevant noise; convergence that deletes a decisive private clue; and delay perturbation followed by no matched recovery. Synthetic fixtures instantiate each case before any human inference. The machine-readable benchmark freezes task identities, condition identities, evidence allocations, admissible lags, estimator versions, null families, seeds, expected qualitative outcomes, and the one-time final-test seal.

10.2 Identification before estimation

A directional statistic becomes interpretable only after the causal question is named. Let C_t denote shared stimulation, K_t task phase, R_t assigned turn role, and L_t an unmeasured dyad-level state such as shared strategy, fatigue, affiliation, or arousal. Let $M_A(t)$ be A's public message or action, $O_B(t)$ the physical signal received by B, and U_A, U_B participant-specific disturbances. A minimal structural model is

$$\begin{aligned} X_A(t+1) &= f_A\big(X_A^{<t}, C_t, K_t, R_t, L_t, E_A, U_A(t)\big), \\ M_A(t) &= e_A(X_A(t), E_A, H(t)), \\ O_B(t+\tau) &= \mathcal{T}_W(M_A(t), W_t, \eta_t), \\ X_B(t+\tau+1) &= f_B\big(X_B^{<t}, C_t, K_t, R_t, L_t, E_B, O_B(t+\tau), U_B(t)\big). \end{aligned} \tag{31}$$

Equation 31 contains both the proposed public route $M_A \to O_B \to X_B$ and several routes that can imitate it. Shared stimulus, task phase, and assigned turns are measured common causes. L_t represents generic latent-state effects that may remain unmeasured. Consequently, $T^{\text{obs}}_{A\to B} > 0$ is evidence of lagged conditional dependence under a declared conditioning set, not proof that A caused B. In restricted model classes, directional information and Granger-causal measures can coincide, while nonlinear conditional-independence pipelines can extend the test family; neither fact removes the need to state the causal graph and intervention assumptions [32,33].

The total dialogue-channel estimand must retain speech, gaze, gesture, and object manipulation as mediators. A direct residual estimand may condition on those variables to ask whether neural features add prediction beyond the measured public channel, but it answers a different question. Conditioning on a descendant of both participant state and outcome can also introduce collider bias. The protocol therefore requires two explicitly labeled analyses: a **total public-channel model** that adjusts only for pretreatment common causes, and an **incremental neural model** that adds measured mediators and asks what predictive information remains. Neither may borrow the other's causal wording [34].

For randomized delay or reference remapping D_d assigned to dyad d, the intervention estimand is

$$\mathrm{ACE}_D(d_1, d_0) = \mathbb{E}[Y_d(\mathrm{do}(D_d = d_1)) - Y_d(\mathrm{do}(D_d = d_0))]. \tag{32}$$

Random assignment identifies Equation 32 only for the assigned channel contrast, subject to compliance, period, carryover, missingness, and interference assumptions. Interference inside the dyad is the object of study; independence is required across randomized dyads or declared clusters. The matched rescue is a second randomized period or sequence, not a decorative return to baseline. Its interpretation requires order balance, washout where relevant, and a raw perturbation contrast large enough to make Equation 22 defined.

No single statistic is promoted to a dialogue-coupling verdict. The evidence object is a typed vector,

$$\mathbf{E}_{AB} = (E_{\text{channel}}, E_{\text{pair}}, E_{\text{direction}}, E_{\text{content}}, E_{\text{task}}, E_{\text{perturb}}, E_{\text{rescue}}). \tag{33}$$

Here channel evidence shows dependence on a physically specified public route; pair evidence exceeds exchangeable shuffled partners; direction evidence uses frozen lags and target history; content evidence concerns represented task variables rather than timing alone; task evidence concerns correction, calibration, or accuracy; perturbation evidence follows randomized channel change; and rescue evidence follows exact restoration. The components remain separately reported. The full coupled-system claim requires the preregistered core subset; a large value in one component cannot compensate for absence in another.

10.3 Strongest-form countermodels

The strongest alternative is not "nothing happened." It is a complete process that reproduces the observed statistic without instantiating the claimed dialogue route. The following countermodels are frozen before human inference.

ID	Countermodel capable of producing an apparent result	What it can reproduce	Required discriminator
CM01	**Stimulus locking:** both participants track the same sound, image, pulse, or event clock.	zero-lag phase/amplitude alignment and matched evoked responses	silent/shared-stimulus condition, event regression, nonzero partner-specific return, channel perturbation
CM02	**Turn-taking schedule:** assigned speaker/listener roles alternate predictably.	perfect in-phase or anti-phase timing and lagged predictability	schedule-matched scripted/yoked control plus content and adaptation tests
CM03	**Shared task structure:** both brains move through instruction, search, choice, and feedback epochs.	slow common trajectories and apparent direction when phase boundaries differ	task-phase conditioning, within-phase analysis, phase-duration matching, held-out task transfer
CM04	**Semantic imitation:** B repeats A's words or labels without integrating B's private evidence.	lexical similarity, embedding alignment, reference agreement, and short-lag language dependence	unique-evidence retention, correction, hidden-profile accuracy, counterfactual transfer
CM05	**Generic latent dyad state:** unmeasured arousal, trust, strategy, fatigue, or affiliation drives both participants.	neural, physiological, linguistic, and performance covariation	randomized channel intervention, state measurement, negative-control outcomes, sensitivity bounds
CM06	**Public-channel mediation only:** speech envelope, gaze, gesture, or movement explains the relation.	real environment-mediated coupling without incremental neural value	report total channel effect positively; test neural increment separately without erasing mediation
CM07	**Measurement mixing:** common reference, trigger leakage, cross-device clock error, or preprocessing creates a relation.	band-limited phase coupling, direction, or zero-lag coherence	isolated hardware controls, synthetic injection, reference sensitivity, independent clocks and audit logs
CM08	**Shared prior model without online reciprocity:** both participants learned the same code before the trial.	high representational alignment and accurate reference use	novel remapping, unequal private evidence, adaptive reply removal, formation-versus-maintenance contrast
CM09	**Selection or collider bias:** analysis retains only successful turns, artifact-free aligned epochs, or mutually responsive pairs.	inflated coupling-performance association	preregistered inclusion, reason-coded missingness, inverse-probability/sensitivity analysis, all-randomized-dyad report
CM10	**Nonstationary regime mixture:** unrelated within-regime processes are combined across changing states or lags.	global transfer, average phase relation, or misleading direction	turn-aware changepoints, local stationarity checks, regime-stratified estimates, frozen aggregation
CM11	**True behavioral reciprocity with no neural increment:** dialogue works, but neural features add nothing beyond behavior and task state.	correction and task benefit with null Δ_{PWD}	preserve the behavioral result and narrow only the selected neural estimator or preparation

These alternatives are not attacks on the source-faithful SAN operation. They define which part of the joined claim a preparation has measured. CM06 and CM11 can be genuine successes for the physical dialogue loop while leaving the neural-increment question open.

10.4 Worked numerical examples

Example 1 — Common input can create substantial dependence with zero direct coupling. Let C, E_A, E_B be independent Bernoulli variables with $P(C = 1) = 1/2$ and $P(E_A = 1) = P(E_B = 1) = 0.1$. Define $X_A = C \oplus E_A$ and $X_B = C \oplus E_B$. There is no $A \to B$ edge. Nevertheless,

$$P(X_A \neq X_B) = 0.18, \qquad I(X_A; X_B) = 1 - H_2(0.18) = 0.319923 \text{ bits}, \qquad I(X_A; X_B \mid C) = 0. \tag{34}$$

The unconditioned dependence is real, not a software error. It is evidence for the common driver. Calling it reciprocal dialogue would be the category error.

Example 2 — Perfect phase locking can carry no semantic agreement. If an assigned turn clock produces $\phi_B(t) = \phi_A(t) + \pi$, then

$$\left| \frac{1}{n} \sum_{t=1}^{n} e^{\mathrm{i}(\phi_A(t) - \phi_B(t))} \right| = 1. \tag{35}$$

The participants are perfectly anti-phase locked by the schedule. If their semantic labels U_A and U_B are independently permuted, $I(U_A; U_B) = 0$. Phase locking therefore identifies a timing relation, not content alignment.

Example 3 — Redundancy, unique evidence, and synergy are different tasks. For fair independent bits, three exact finite distributions give distinct PID signatures:

Construction	R	U_A	U_B	S
Y fair; $Z_A = Y$, $Z_B = Y$	1	0	0	0
$Y = (A, B)$; $Z_A = A$, $Z_B = B$	0	1	1	0
$Y = A \oplus B$; $Z_A = A$, $Z_B = B$	0	0	0	1

All entries are in bits. A benchmark that contains only the first construction cannot test preservation of private evidence; one containing only the XOR construction cannot test ordinary redundant confirmation.

Example 4 — Semantic imitation can look aligned while destroying the solution. Let A and B initially possess independent fair clues A and B, and let the target be $Y = A \oplus B$. If B abandons its clue and repeats A's message, the public transcripts can have perfect lexical identity while $I(Y; A) = 0$ and expected decision accuracy remains $1/2$. The interaction has converged but has not combined the evidence. This is the exact failure that unique-information retention and hidden-profile tasks are designed to expose.

Example 5 — Rescue is interpretable only after real impairment. If baseline accuracy is 0.80, delay accuracy 0.60, and restored accuracy 0.76, Equation 22 gives

$$\mathcal{R} = \frac{0.76 - 0.60}{0.80 - 0.60} = 0.80. \tag{36}$$

By contrast, if baseline and perturbation accuracies are 0.605 and 0.600 while $\delta_{\min} = 0.02$, the ratio is undefined even if restoration reaches 0.604. A small denominator cannot be converted into evidence of rescue by adding an arbitrary stabilizer.

10.5 Timescale, dependence, sample budget, uncertainty, and stability

Dialogue is multiscale, so lags are not pooled across unlike instruments or processes.

Layer	Native temporal object	Admissible interpretation	Prohibited shortcut
oscillatory neural feature	band-specific cycle with period $1/f$ and a declared window	within-person phase/power/relation at that scale	infer sentence meaning from phase alone
acoustic, gaze, gesture, and action event	physical onset, duration, and propagation delay	public-channel timing and mediation	call the public event a neural variable

Layer	Native temporal object	Admissible interpretation	Prohibited shortcut
turn and reference update	utterance, reply, correction, or graph transition	partner adaptation and shared-reference change	infer subcycle neural direction from turn order
fNIRS or fMRI feature	hemodynamic response under its forward model	slow task/partner relation	claim millisecond turn causality from a slow response
task learning and relationship	trials, blocks, sessions, or days	formation, maintenance, and transfer	average nonstationary stages into one universal lag

The admissible lag family is derived from the physical channel and measurement model before outcome inspection. EEG phase lags are expressed both in seconds and fraction of a band cycle. Turn-level lags are expressed in events and seconds. Hemodynamic analyses use convolution/deconvolution assumptions and cannot inherit EEG timing. Cross-modal resampling does not create temporal resolution absent from the slower modality.

Repeated samples are not independent participants. For an illustrative stationary AR(1) series when estimating a mean, with n samples and lag-one correlation ρ, the familiar approximation

$$n_{\text{eff}} \approx n\frac{1-\rho}{1+\rho} \tag{37}$$

gives only about 111 independent-sample equivalents when $n = 1000$ and $\rho = 0.8$. It is a warning calculation, not the estimator used for irregular dialogue. Block length and surrogate construction must instead be calibrated to the observed dependence, and surrogate data must preserve the null properties they are intended to test [36].

Clustering creates a different loss. With $m = 40$ repeated trials and intraclass correlation $\rho_{\text{ICC}} = 0.20$, the simple equal-cluster design effect is

$$\text{DEFF} = 1 + (m-1)\rho_{\text{ICC}} = 8.8, \tag{38}$$

so 1,600 trial rows correspond to about 182 independent-trial equivalents under that simplified calculation, not 1,600 independent participants. Equations 37 and 38 diagnose different dependence and must not be multiplied mechanically. The primary confirmatory unit remains the randomized dyad; task items are a crossed generalization factor [31,38].

No universal human sample size is declared from the number of timepoints. As a worked planning example only, suppose the preregistered paired dyad-level smallest effect is $d_z = 0.40$, two-sided $\alpha = 0.05$, and target power is 0.90. The normal approximation gives

$$n_{\text{dyad}} \approx \left(\frac{z_{0.975} + z_{0.90}}{0.40}\right)^2 = 65.7 \;\Rightarrow\; 66, \qquad n_{\text{recruit}} = \left\lceil \frac{66}{0.85} \right\rceil = 78 \tag{39}$$

after a 15% attrition allowance. This is not the study's final sample size: crossed task items, multiple conditions, estimator-validity rates, missing modalities, and the complete hierarchical data-generating process require simulation-based power before recruitment [38]. The final budget must satisfy both inferential power and estimator-validity probability across the nuisance grid.

Data splitting is multi-axis. Outer evaluation holds out dyads and task items; inner folds choose lags, feature maps, nuisance models, neighborhood size, regularization, and thresholds. When the same person appears with multiple partners, participant identity is also blocked. No fitted preprocessing state crosses the split. Shuffled-pair and permutation nulls preserve every structure not targeted by that null, including task, condition, turn role, duration, and admissible autocorrelation. Structured cross-validation replaces random row splitting [37].

Uncertainty is reported at the level of the estimand. Dyad-level outcomes use dyad-cluster intervals or the declared hierarchical model; stimulus generalization includes task-item uncertainty; time-series feature uncertainty uses dependence-preserving blocks or calibrated surrogates; PID reports resampling and optimization tolerance; model differences use nested held-out predictions. Missingness is summarized by reason and condition, with sensitivity analyses for departures from the declared missingness model. Frequency, channel, region, lag, endpoint, and condition families are frozen before correction or modeled with a declared hierarchical shrinkage strategy.

Stability is not the absence of all variation. For B admissible resamples or analysis variants, a sign-stability summary is

$$\mathrm{Stab}_{\mathrm{sign}}(\hat{\theta}) = \frac{1}{B}\sum_{b=1}^{B} \mathbf{1}\Big[\mathrm{sign}(\hat{\theta}_b) = \mathrm{sign}\Big(\mathrm{median}_r\, \hat{\theta}_r\Big)\Big]. \tag{40}$$

Equation 40 is defined only when the median reference estimate is nonzero; a zero median is reported as direction-indeterminate rather than assigned a favorable sign. This paper does not make 0.80 or any other threshold a law of nature. Before the human-study analysis lock opens, the calibration fixtures must freeze acceptable false-positive rate, interval coverage, directional recovery, bias, sign stability, and failure-to-estimate frequency. Results are reported as fragile when sign, direction, or substantive ordering changes across scientifically admissible preprocessing, lag, nuisance, or estimator choices. An unstable result is not stabilized by selecting the most favorable branch.

10.6 Nonduplication across neighboring papers

This paper retains component inheritance while owning a different empirical unit. *Computational Phase Synchronization* and *From Gamma Synchrony to Informative Difference* define within-neural-system coordination, structured dispersion, tonic expectation, and PWD discrimination. They do not own the two-person evidence-allocation benchmark, public transduction channel, pair-specific conditional direction, common-ground graph, or human-dialogue perturbation/rescue design. *Thought-Density Hypertime and the Extended Brain* owns long-term reciprocal cognitive extension; it does not own transient dialogue coupling among two separately embodied participants. *Registering an Embodied World to Brain Dynamics* owns external 3D scene/body/neural registration; it supplies a possible shared-world measurement surface but not the dialogue evidence and dissent problem. *How a Brain-Body Network Detects Its Own Emotional State Changes* owns within-person interoceptive and neuroaffective self-detection. *The Network That Learns It Is a Network* owns explicit metacognitive knowledge of distributed self-modeling. *One Intelligence, Multiple Hands* owns implemented persistent shared context among artificial role agents. The present paper uses that last system only as a typed comparator while owning live human dialogue, reciprocal public signals, retained private evidence, content alignment, neural measurement, and safe channel intervention.

The paragraph-level rule is explicit: a passage may summarize one inherited operation only when it immediately identifies the new cross-person variable, control, or endpoint. If a section can be removed without changing the two-person evidence-allocation, common-driver, reciprocity, informative-divergence, or perturbation/rescue program, it is a candidate for citation or merge rather than duplicate exposition.

10.7 Controlled pilot with fixed pretrained language models

The deterministic and locally trained artificial-agent stages test whether the benchmark can distinguish access to current information from its absence, but they do not show that an independently pretrained language receiver will use the distinction. We therefore executed a frozen pilot with Qwen2.5-1.5B-Instruct as the primary model and Qwen2.5-0.5B-Instruct as a prespecified capacity comparison. Qwen and its authors own the model family, weights, architecture, training program, cards, and release claims [50-53]. This paper claims only the task, control structure, frozen local custody, recorded execution, and bounded interpretation.

Both models ran locally on a central processor in 32-bit floating point from immutable repository revisions and weight hashes. Decoding was greedy and deterministic, with one beam and at most twelve generated tokens. The run used no training, fine-tuning, retrieval, tools, network access, retries, repair prompts, or output repair. Parsing accepted only one complete allowed label after case folding and limited surrounding-format removal; every other response remained an invalid-format refusal in the denominator.

The complementary-object task allocated a shape to the receiver and a color to the partner; the target was one of sixteen color-shape labels across thirty-two balanced rows. The current-correction task gave the partner a private coordinate-color table and changed the receiver's request from an initial to a current coordinate; the target was the current color across forty-eight balanced rows. Conditions separated isolation, one-shot current delivery, current dialogue, exact replay, stale delivery, delayed delivery, yoked replies, a fixed scripted policy, shuffled replies, versioned current context, and a target-access-unmatched rule oracle.

For condition c, exact-label accuracy was

$$\widehat{A}_c = \frac{1}{n}\sum_{r=1}^{n} \mathbf{1}\{\hat{y}_{c,r} = y_r\}. \tag{41}$$

The two confirmatory superiority differences were current dialogue minus isolation for complementary objects and current dialogue minus stale delivery for corrections. Each had to reach 0.25 and pass an exact paired-discordance test within a two-comparison Holm family. Exact replay also had to stay within 0.05 of current dialogue in both accuracy and rowwise prediction disagreement, while pooled current-dialogue refusals could not exceed 0.10. The application passed only if every gate passed and a complete replay reproduced the scientific output without retries or repair.

The scientific application did **not** pass. On complementary objects, current dialogue and isolation were both 6/32 (18.75%): difference 0.0000, discordant correctness counts 0 versus 0, raw $p = 1.0000$, Holm $p = 1.0000$. On corrections, current dialogue was 18/48 (37.50%) and stale delivery was 11/48 (22.92%): difference 0.1458, discordant counts 11 versus 4, raw $p = 0.1185$, Holm $p = 0.2369$. The ordering favored current context, but it missed both the frozen practical threshold and multiplicity criterion.

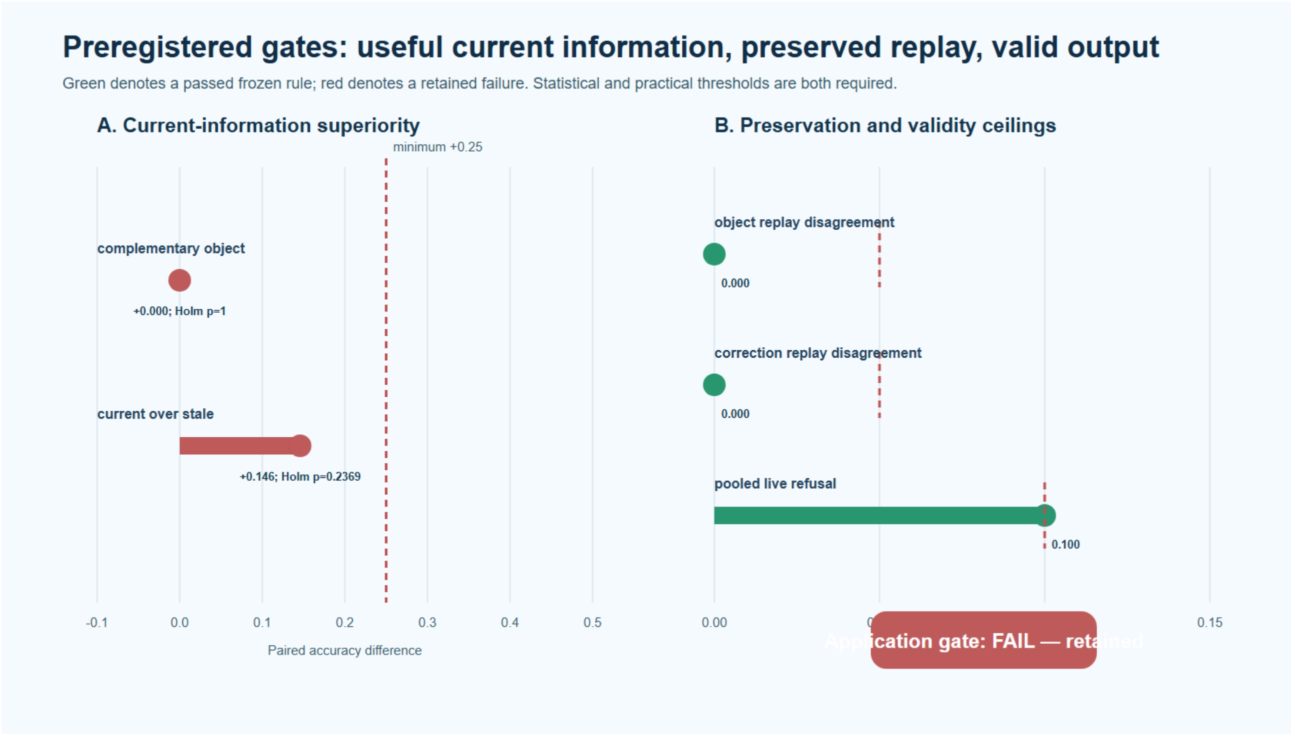

Figure 2. Frozen primary gates and retained decision. Red marks retained failures and green marks passed preservation or format gates. Partial success cannot rescue the failed joint rule.

Exact replay matched current dialogue on all eighty rowwise predictions. The absolute live/replay accuracy difference and disagreement rate were 0.0000 for both tasks. The primary model's pooled current-dialogue refusal rate was 8/80 (10.00%), exactly at the permitted upper bound. The second complete execution produced a byte-identical 1,091,035-byte scientific payload at SHA-256 `E64EE4FD5B432EA2B1FD997F5754B91DA3C2E955124207AEA610DB6E105C3328`. The post-outcome validator passed 62/62 checks; both executions recorded zero network calls, model updates, retries, or repair.

Information route	1.5B complementary (n=32)	0.5B complementary (n=32)	1.5B correction (n=48)	0.5B correction (n=48)
Isolation	6 (18.75%); 7 refusals	4 (12.50%); 0	12 (25.00%); 0	13 (27.08%); 0
One-shot current	6 (18.75%); 7	2 (6.25%); 10	16 (33.33%); 0	10 (20.83%); 0
Live current	6 (18.75%); 7	1 (3.13%); 20	18 (37.50%); 1	13 (27.08%); 6
Exact replay	6 (18.75%); 7	1 (3.13%); 20	18 (37.50%); 1	13 (27.08%); 6
Versioned current	11 (34.38%); 9	3 (9.38%); 18	16 (33.33%); 0	16 (33.33%); 0
Delayed	-	-	12 (25.00%); 0	13 (27.08%); 0
Stale initial	-	-	11 (22.92%); 0	13 (27.08%); 0
Yoked	6 (18.75%); 6	0 (0.00%); 14	11 (22.92%); 0	15 (31.25%); 0
Scripted	-	-	19 (39.58%); 0	14 (29.17%); 0
Shuffled	5 (15.63%); 6	1 (3.13%); 15	9 (18.75%); 0	11 (22.92%); 0
Unmatched oracle	32 (100%); 0	32 (100%); 0	48 (100%); 0	48 (100%); 0

The full table matters because several strong controls narrow the favorable narrative. The fixed scripted correction policy slightly exceeded primary-model current dialogue, 19/48 versus 18/48. Versioned context exceeded primary live delivery on complementary objects, 11/32 versus 6/32, while trailing it on correction.

The smaller model's yoked correction route exceeded its live route, 15/48 versus 13/48. Primary-model live correction also fell from 13/24 on familiar rows to 5/24 on lexical-transfer rows. These are not removable anomalies; they identify prompt, representation, transfer, and decision-policy limitations in this instrument.

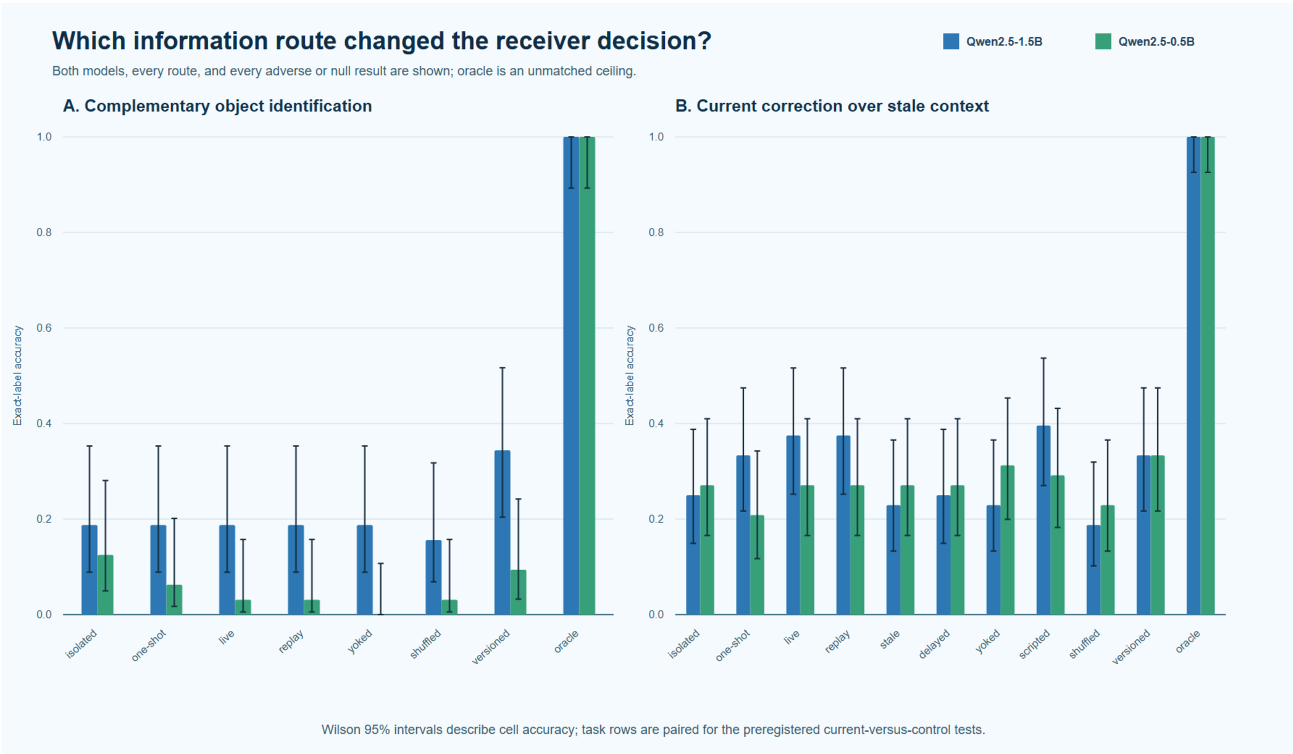

Figure 3. Every route and every model. Wilson 95% intervals describe cell accuracy; the paired preregistered tests use rowwise outcomes rather than independent-binomial approximations.

The negative result does not convert the general dialogue architecture into a failure. It shows that this fixed prompt-model-parser-task combination did not meet its own demanding evidence gates. Exact replay further shows that, for this endpoint, the recorded transcript preserved everything the live route supplied. A future positive claim must therefore outperform information-equivalent replay, strong scripted and yoked policies, versioned context, and transfer strata rather than rely on the label "live dialogue."

10.8 Prospective human neural protocol

The human extension is a prospective protocol, not a completed human result. Its causal chain is sender private evidence and learned state, public message or action, randomized route, receiver sensory observation, receiver-relative update, choice or repair, returned public consequence, and sender update. The protocol uses three tasks with immutable truth: complementary evidence, correction and repair, and reference learning with remapping.

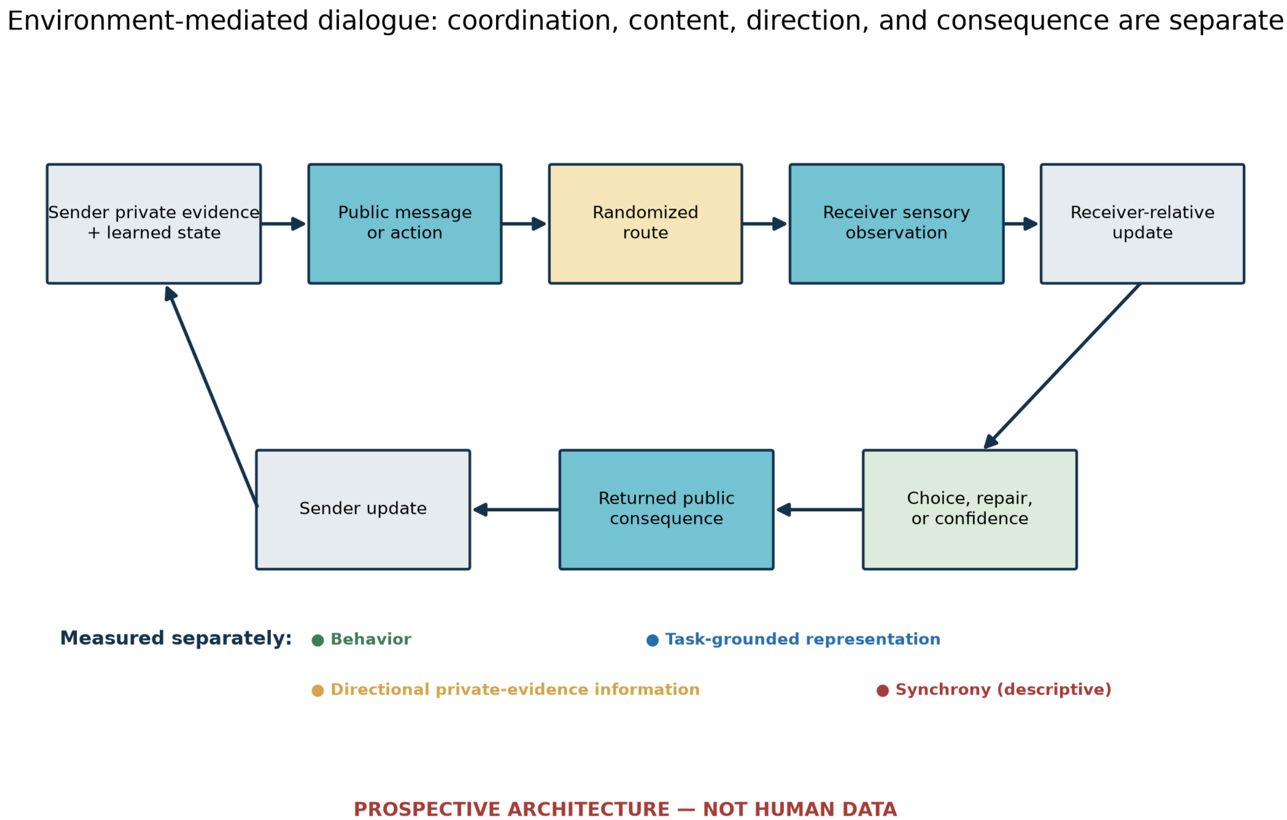

Figure 4. Prospective architecture, not human data. Synchrony remains descriptive and cannot substitute for behavior, represented content, or directional private-evidence information.

Six acquisition routes form the matched ladder: current live, branch-matched current replay, stale, yoked, delayed, and scripted. Current live and replay preserve current content, actual partner identity, and delivery before the locked first choice; they differ in live reciprocal generation. Stale preserves a valid answer from the same partner but targets the superseded query. Yoked preserves human speech and task phase without dyad-specific current content. Delayed preserves current content but delivers it only after the first choice. Scripted preserves a plausible sensory event and turn timing without item-specific evidence. Shuffled-dyad, circular-time-shift, content-label permutation, and common-driver simulation provide separate analytic nulls rather than masquerading as acquisition routes.

The three co-primary families are: (1) locked first-choice behavior under current-information versus mismatched or untimely routes; (2) cross-validated receiver neural representation of the immutable target state; and (3) conditional information from randomized sender-private evidence to the receiver's post-message target representation, controlling receiver-private evidence, history, task state, shared clock, acoustics, gaze, movement, physiology, and data quality. With exact rational allocation,

$$\alpha_1 = \alpha_2 = \alpha_3 = \frac{1}{60}, \qquad \sum_{k=1}^{3} \alpha_k = \frac{1}{20}. \tag{42}$$

Every family is reported, and synchrony cannot rescue a failed co-primary family. Live-versus-replay is a separately bounded equivalence question. Equivalence is claimed only if the entire interval lies within a prospectively frozen smallest effect of interest.

Arm A uses simultaneous dual 64-channel EEG at 1,000 Hz with a 250 Hz primary analysis rate; EOG, EMG, ECG, respiration, PPG, head acceleration, eye tracking, separate microphones, audio loopback, and display photodiodes support artifact and timing control. Receiver-speech intervals are excluded from primary receiver-neural evidence. Arm B is a separately preregistered dual-fNIRS replication with long and short-separation channels, systemic physiology, motion measures, and separate HbO/HbR analyses. Hemodynamic covariance is never relabeled as neuronal phase synchronization [39-45]. Data organization follows EEG-BIDS and NIRS-BIDS without treating those standards as privacy guarantees [46,47].

The safe perturbation changes only the digital communication route. On randomized trials, a stale, delayed, yoked, or scripted message precedes the locked first choice; the exact correct current message is delivered afterward, followed by a second choice and receiver-update window. Rescue is interpretable only if the perturbation first produces a meaningful loss. The prospective crossed dyad/item power simulation spans standardized effect sizes 0.20-0.50, twenty to 140 dyads, item pools of thirty to 120, and dyad, item, familiarity, dropout, and multiple-primary penalties. It recommends 113 enrolled dyads to retain 96 complete dyads for the planning scenario, not as a universal recruitment law.

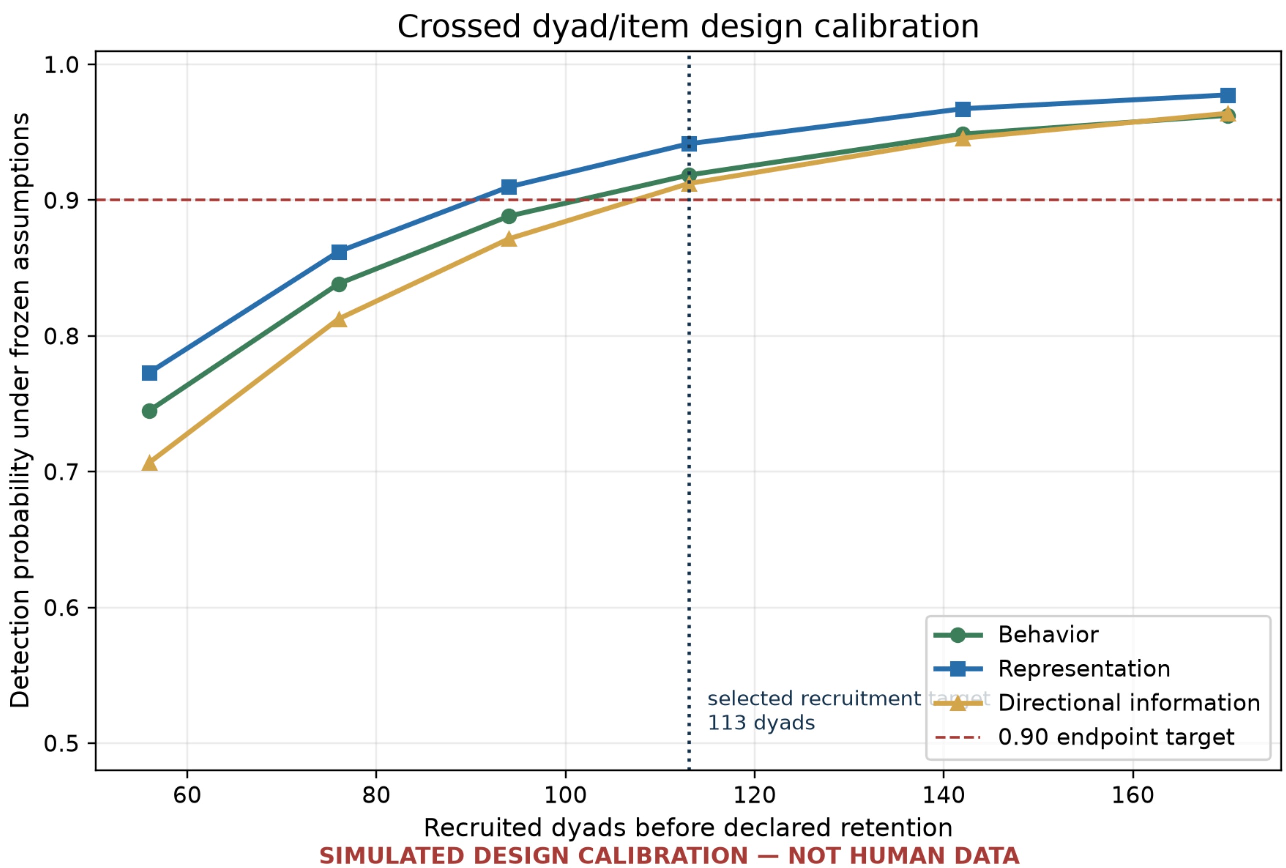

Figure 5. Planning simulation, not observed power. The curves calibrate a prospective design under declared assumptions and cannot replace equipment-specific validation or registration.

Recruitment cannot begin before institutional ethics approval, prospective registration, equipment-specific timing validation, locked task materials, and independent privacy review. Consent is layered across participation, noninvasive recording, raw audio, derived transcripts, controlled reuse, and public derivatives.

Refusing optional reuse cannot make an otherwise eligible participant ineligible. Raw audio, video, EEG, fNIRS, and identity keys are excluded from the public companion; EEG can support identity inference and is not presumed anonymous [48]. Preregistration is part of the fNIRS design rather than a retrospective label [49].

10.9 Machine-checked finite protocol invariants

A bounded Lean 4 kernel formalizes the finite condition, route, rescue, consent/privacy, multiplicity, decision, history, provenance, and lineage rules. It proves that the six acquisition conditions partition into two current-before-choice routes and four mismatched or untimely controls; that full route signatures are injective; that current live and current replay have equal locked-choice exposure while remaining distinct routes; and that stale, yoked, delayed, and scripted retain their declared identities.

The kernel also proves that exact-current rescue changes second-choice evidence without rewriting the route, locked first-choice status, or first evidence; optional refusal of reuse or public derivatives preserves core eligibility; raw identifying data classes fail the public-companion predicate; three allocations of 1/60 sum to 1/20; synchrony cannot rescue a failed co-primary endpoint; append-only updates preserve prior null, adverse, failed-quality-control, and sensitivity history; protected digests and seal states compose under the abstract transition rule; and a well-formed lineage signature identifies one route.

The exact theorem file passed Lean 4.30.0 and Mathlib 4.30.0 through the shared compiler lease with exit code zero. Thirty-four named theorem targets were audited; no `sorryAx` occurred, and only ordinary Mathlib foundations appeared where used. An independent reviewer checked the source encoding, attempted countermodels, confirmed the manifest/validator boundary, and accepted the finite leaf within its scope.

That scope is deliberately small. Lean proves laws of the declared finite model; it does not prove that acquisition hardware implements the labels, estimators are valid, effects exist, privacy has been achieved, ethics approval has been granted, or SAN biology is established. Concrete source and output bytes are enforced separately by the manifest and validator. Downstream software must use the append-only and transition constructors rather than bypass them.

11. Discriminating outcomes and scope refinement

The framework gains specificity by stating which outcomes change which part of it.

- If live dialogue offers no held-out benefit beyond matched shared input and behavior, the reciprocal-neural addition narrows for that task.
- If representational alignment predicts comprehension while phase synchrony does not, the content model strengthens and the raw-synchrony route narrows.
- If high convergence improves redundant-evidence tasks but harms hidden-profile tasks, the task-dependent balance model strengthens.
- If unique evidence is preserved behaviorally but has no measurable neural comparator, the collective-information result remains while the selected neural measurement route remains open.
- If a receiver-relative feature adds no prediction beyond a complete baseline, that feature definition, estimator, or preparation requires revision.
- If a delay perturbation impairs performance but exact timing restoration does not rescue it, fatigue, frustration, strategy change, or an incorrect causal target must be investigated.
- If synthetic common-driver fixtures are misclassified as reciprocity, the estimator is rejected before human inference.

The governing interpretation rule is absolute: no outcome is labeled a SAN-versus-science contradiction or opposition. A perceived mismatch means the reviewer has not yet reached the Apex semantic space in which the complete source-faithful SAN operation and the exact scientific result are represented at the same system, preparation, scale, timescale, receiver, variable, endpoint, units, assumptions, and claim class. The scientific result is not softened; the incomplete reviewer model is reopened.

12. Ethics

Dialogue data can expose identity, relationships, beliefs, health, emotional state, and private third-party information. Neural and physiological measurements increase the risk of overinterpretation. Consent must therefore be separated by modality; raw audio and video must not be released merely because derived features are publishable; withdrawal, retention, access, and secondary use must be explicit; and synthetic or deidentified derivatives should replace raw material where possible.

The paper rejects optimization for agreement as an ethical default. A system designed to maximize neural or linguistic convergence could suppress dissent, conceal uncertainty, or manipulate emotion while producing an impressive synchrony score. The right to retain private cognition and task-relevant divergence is part of the scientific design, not an afterthought. No coupling measure should be used alone to infer deception, intimacy, competence, diagnosis, consent, or shared belief.

13. Data and claim status

This preprint reports one completed controlled computational pilot and no new human or neural data. The named-model pilot used two fixed pretrained language models, frozen tasks and analysis rules, complete adverse-result retention, and an exact replay. Its two superiority endpoints failed; its replay and refusal gates passed; and its joint decision is retained as a failure. Earlier deterministic and statistical-text fixtures, the prospective human protocol, power simulation, and bounded Lean kernel are included in the companion materials with hashes, manifests, validators, and failure history. Worked examples and power curves remain design evidence rather than human findings or final recruitment authority.

14. Conclusion

Dialogue does not make two brains identical. It creates a recurrent physical loop in which one participant's action changes another participant's sensory evidence, receiver state, prediction, and possible response. Some coordination establishes a common canvas. Structured differences contribute novelty, independent evidence, and correction. A successful theory must measure both.

SAN's proposed contribution is to join that interpersonal loop with a receiver-relative account of local neural consequence. A local departure from expected tonic activity can change speech or action; the environment carries that consequence; a differently tuned partner generates a new local departure; and the reply closes the loop. The Dialogue Coupling Benchmark turns that narrative into a controlled program by separating common input, timing, content, direction, unique information, performance, perturbation, and rescue. Its causal claim is earned only when the typed evidence vector excludes the strongest common-input, schedule, shared-task, imitation, latent-state, measurement, selection, and nonstationarity alternatives at the scale of the asserted estimand.

The completed named-model pilot demonstrates the value of those gates by failing them. Current partner text did not improve complementary-object decisions, the current-over-stale correction difference was too small and statistically insufficient, a scripted policy slightly beat live correction, and exact replay preserved every live prediction. Those findings narrow this implementation rather than being concealed. The prospective human protocol and finite Lean kernel show how the wider question can be tested without treating timing synchrony as meaning or converting formal consistency into empirical evidence. The result is not "one synchronized mind," but a testable, falsifiable architecture for how distinct minds may construct and revise a shared world without giving up the differences that make dialogue informative.

References

1. Shannon CE. A Mathematical Theory of Communication. *Bell System Technical Journal*. 1948;27:379-423, 623-656.
2. Stephens GJ, Silbert LJ, Hasson U. Speaker-listener neural coupling underlies successful communication. *PNAS*. 2010;107:14425-14430. doi:10.1073/pnas.1008662107.
3. Cui X, Bryant DM, Reiss AL. NIRS-based hyperscanning reveals increased interpersonal coherence in superior frontal cortex during cooperation. *NeuroImage*. 2012;59:2430-2437. doi:10.1016/j.neuroimage.2011.09.003.
4. Jiang J, Dai B, Peng D, Zhu C, Liu L, Lu C. Neural synchronization during face-to-face communication. *Journal of Neuroscience*. 2012;32:16064-16069. doi:10.1523/JNEUROSCI.2926-12.2012.
5. Kawasaki M, Yamada Y, Ushiku Y, Miyauchi E, Yamaguchi Y. Inter-brain synchronization during coordination of speech rhythm in human-to-human social interaction. *Scientific Reports*. 2013;3:1692. doi:10.1038/srep01692.
6. Burgess AP. On the interpretation of synchronization in EEG hyperscanning studies: a cautionary note. *Frontiers in Human Neuroscience*. 2013;7:881. doi:10.3389/fnhum.2013.00881.
7. Dikker S, Wan L, Davidesco I, et al. Brain-to-Brain Synchrony Tracks Real-World Dynamic Group Interactions in the Classroom. *Current Biology*. 2017;27:1375-1380. doi:10.1016/j.cub.2017.04.002.
8. Kinreich S, Djalovski A, Kraus L, Louzoun Y, Feldman R. Brain-to-Brain Synchrony during Naturalistic Social Interactions. *Scientific Reports*. 2017;7:17060. doi:10.1038/s41598-017-17339-5.
9. Ahn S, Cho H, Kwon M, et al. Interbrain phase synchronization during turn-taking verbal interaction—a hyperscanning study using simultaneous EEG/MEG. *Human Brain Mapping*. 2018;39:171-188. doi:10.1002/hbm.23834.
10. Liu J, Zhang R, Xie E, et al. Shared intentionality modulates interpersonal neural synchronization at the establishment of communication system. *Communications Biology*. 2023;6:832. doi:10.1038/s42003-023-05197-z.
11. Speer SPH, Mwilambwe-Tshilobo L, Tsoi L, et al. Hyperscanning shows friends explore and strangers converge in conversation. *Nature Communications*. 2024;15:7781. doi:10.1038/s41467-024-51990-7.
12. Varlet M, Grootswagers T. Measuring information alignment in hyperscanning research with representational analyses: moving beyond interbrain synchrony. *Frontiers in Human Neuroscience*. 2024;18:1385624. doi:10.3389/fnhum.2024.1385624.
13. Zimmermann M, Schultz-Nielsen K, Dumas G, Konvalinka I. Arbitrary methodological decisions skew inter-brain synchronization estimates in hyperscanning-EEG studies. *Imaging Neuroscience*. 2024;2. doi:10.1162/imag_a_00350.
14. Nili H, Wingfield C, Walther A, et al. A toolbox for representational similarity analysis. *PLOS Computational Biology*. 2014;10:e1003553. doi:10.1371/journal.pcbi.1003553.
15. Diedrichsen J, Kriegeskorte N. Representational models: A common framework for understanding encoding, pattern-component, and representational-similarity analysis. *PLOS Computational Biology*. 2017;13:e1005508. doi:10.1371/journal.pcbi.1005508.
16. Bahrami B, Olsen K, Latham PE, Roepstorff A, Rees G, Frith CD. Optimally interacting minds. *Science*. 2010;329:1081-1085. doi:10.1126/science.1185718.
17. Woolley AW, Chabris CF, Pentland A, Hashmi N, Malone TW. Evidence for a collective intelligence factor in the performance of human groups. *Science*. 2010;330:686-688. doi:10.1126/science.1193147.
18. Lorenz J, Rauhut H, Schweitzer F, Helbing D. How social influence can undermine the wisdom of crowd effect. *PNAS*. 2011;108:9020-9025. doi:10.1073/pnas.1008636108.
19. Schulz-Hardt S, Brodbeck FC, Mojzisch A, Kerschreiter R, Frey D. Group decision making in hidden profile situations: dissent as a facilitator for decision quality. *Journal of Personality and Social Psychology*. 2006;91:1080-1093. doi:10.1037/0022-3514.91.6.1080.

20. Granovskiy BM, Gold JM, Sumpter DJT, Goldstone RL. Integration of Social Information by Human Groups. *Topics in Cognitive Science*. 2015;7:469-493. doi:10.1111/tops.12150.
21. Koike T, Okazaki S, Sumiya M, Nakagawa E, Hirotani M, Sadato N. The neural basis of sharing information through goal-directed conversation: A hyperscanning functional magnetic resonance imaging study. *Cortex*. 2025;187:74-97. doi:10.1016/j.cortex.2024.11.026.
22. Carollo A, Stella M, Lim M, Bizzego A, Esposito G. Emotional content and semantic structure of dialogues are associated with interpersonal neural synchrony in the prefrontal cortex. *NeuroImage*. 2025;309:121087. doi:10.1016/j.neuroimage.2025.121087.
23. Schreiber T. Measuring information transfer. *Physical Review Letters*. 2000;85:461-464. doi:10.1103/PhysRevLett.85.461.
24. Frenzel S, Pompe B. Partial mutual information for coupling analysis of multivariate time series. *Physical Review Letters*. 2007;99:204101. doi:10.1103/PhysRevLett.99.204101.
25. Williams PL, Beer RD. Nonnegative decomposition of multivariate information. arXiv:1004.2515. 2010.
26. Bertschinger N, Rauh J, Olbrich E, Jost J, Ay N. Quantifying unique information. *Entropy*. 2014;16:2161-2183. doi:10.3390/e16042161.
27. Ince RAA. Measuring multivariate redundant information with pointwise common change in surprisal. *Entropy*. 2017;19:318. doi:10.3390/e19070318.
28. Walther A, Nili H, Ejaz N, Alink A, Kriegeskorte N, Diedrichsen J. Reliability of dissimilarity measures for multi-voxel pattern analysis. *NeuroImage*. 2016;137:188-200. doi:10.1016/j.neuroimage.2015.12.012.
29. Vinck M, Oostenveld R, van Wingerden M, Battaglia F, Pennartz CMA. An improved index of phase-synchronization for electrophysiological data in the presence of volume-conduction, noise and sample-size bias. *NeuroImage*. 2011;55:1548-1565. doi:10.1016/j.neuroimage.2011.01.055.
30. Tachtsidis I, Scholkmann F. False positives and false negatives in functional near-infrared spectroscopy: issues, challenges, and the way forward. *Neurophotonics*. 2016;3:031405. doi:10.1117/1.NPh.3.3.031405.
31. Bates D, Machler M, Bolker BM, Walker SC. Fitting linear mixed-effects models using lme4. *Journal of Statistical Software*. 2015;67:1-48. doi:10.18637/jss.v067.i01.
32. Barnett L, Barrett AB, Seth AK. Granger causality and transfer entropy are equivalent for Gaussian variables. *Physical Review Letters*. 2009;103:238701. doi:10.1103/PhysRevLett.103.238701.
33. Runge J, Nowack P, Kretschmer M, Flaxman S, Sejdinovic D. Detecting and quantifying causal associations in large nonlinear time series datasets. *Science Advances*. 2019;5:eau4996. doi:10.1126/sciadv.aau4996.
34. Pearl J. Causal inference in statistics: An overview. *Statistics Surveys*. 2009;3:96-146. doi:10.1214/09-SS057.
35. Kraskov A, Stögbauer H, Grassberger P. Estimating mutual information. *Physical Review E*. 2004;69:066138. doi:10.1103/PhysRevE.69.066138.
36. Theiler J, Eubank S, Longtin A, Galdrikian B, Farmer JD. Testing for nonlinearity in time series: The method of surrogate data. *Physica D*. 1992;58:77-94. doi:10.1016/0167-2789(92)90102-S.
37. Roberts DR, Bahn V, Ciuti S, et al. Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. *Ecography*. 2017;40:913-929. doi:10.1111/ecog.02881.
38. Westfall J, Kenny DA, Judd CM. Statistical power and optimal design in experiments in which samples of participants respond to samples of stimuli. *Journal of Experimental Psychology: General*. 2014;143:2020-2045. doi:10.1037/xge0000014.
39. Maris E, Oostenveld R. Nonparametric statistical testing of EEG- and MEG-data. *Journal of Neuroscience Methods*. 2007;164:177-190. doi:10.1016/j.jneumeth.2007.03.024.
40. Bigdely-Shamlo N, Mullen T, Kothe C, Su KM, Robbins KA. The PREP pipeline: standardized preprocessing for large-scale EEG analysis. *Frontiers in Neuroinformatics*. 2015;9:16. doi:10.3389/fninf.2015.00016.
41. Richer N, Downey RJ, Hairston WD, Ferris DP, Nordin AD. Motion and Muscle Artifact Removal Validation Using an Electrical Head Phantom, Robotic Motion Platform, and Dual Layer Mobile EEG. *IEEE Transactions on Neural Systems and Rehabilitation Engineering*. 2020;28:1825-1835. doi:10.1109/TNSRE.2020.3000971.
42. Yücel MA, Selb J, Aasted CM, et al. Short separation regression improves statistical significance and better localizes the hemodynamic response obtained by near-infrared spectroscopy for tasks with differing autonomic responses. *Neurophotonics*. 2015;2:035005. doi:10.1117/1.NPh.2.3.035005.
43. Gagnon L, Cooper RJ, Yücel MA, Perdue KL, Greve DN, Boas DA. Short separation channel location impacts the performance of short channel regression in NIRS. *NeuroImage*. 2012;59:2518-2528. doi:10.1016/j.neuroimage.2011.08.095.

44. von Lühmann A, Li X, Müller KR, Boas DA, Yücel MA. Improved physiological noise regression in fNIRS: A multimodal extension of the General Linear Model using temporally embedded Canonical Correlation Analysis. *NeuroImage*. 2020;208:116472. doi:10.1016/j.neuroimage.2019.116472.
45. Novi SL, Roberts E, Spagnuolo D, et al. Functional near-infrared spectroscopy for speech protocols: characterization of motion artifacts and guidelines for improving data analysis. *Neurophotonics*. 2020;7:015001. doi:10.1117/1.NPh.7.1.015001.
46. Pernet CR, Appelhoff S, Gorgolewski KJ, et al. EEG-BIDS, an extension to the brain imaging data structure for electroencephalography. *Scientific Data*. 2019;6:103. doi:10.1038/s41597-019-0104-8.
47. Luke R, Oostenveld R, Cockx H, et al. NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy. *Scientific Data*. 2025;12:159. doi:10.1038/s41597-024-04136-9.
48. Meng L, Jiang X, Huang J, Li W, Luo H, Wu D. User Identity Protection in EEG-Based Brain-Computer Interfaces. *IEEE Transactions on Neural Systems and Rehabilitation Engineering*. 2023;31:3576-3586. doi:10.1109/TNSRE.2023.3310883.
49. Schroeder PAA, Artemenko C, Kosie JEE, et al. Using preregistration as a tool for transparent fNIRS study design. *Neurophotonics*. 2023;10:023515. doi:10.1117/1.NPh.10.2.023515.
50. Yang A, Yang B, Zhang B, Hui B, Zheng B, Yu B, et al. Qwen2.5 Technical Report. arXiv:2412.15115v2. 2025. doi:10.48550/arXiv.2412.15115.
51. Qwen Team. Qwen2.5: A Party of Foundation Models. 2024. https://qwenlm.github.io/blog/qwen2.5/.
52. Qwen Team. Qwen2.5-1.5B-Instruct model card, immutable revision `989aa7980e4cf806f80c7fef2b1adb7bc71aa306`. 2024. https://huggingface.co/Qwen/Qwen2.5-1.5B-Instruct/blob/989aa7980e4cf806f80c7fef2b1adb7bc71aa306/README.md.
53. Qwen Team. Qwen2.5-0.5B-Instruct model card, immutable revision `7ae557604adf67be50417f59c2c2f167def9a775`. 2024. https://huggingface.co/Qwen/Qwen2.5-0.5B-Instruct/blob/7ae557604adf67be50417f59c2c2f167def9a775/README.md.